

# CareConnect

## Accessibility Requirements

SWEN 661 Team 1

CareConnect is designed for patients with Severe Carpal Tunnel Syndrome (CTS). Standard device interaction - typing, repeated tapping, sustained grip - compresses the median nerve and causes pain. The following requirements ensure every interaction is safe and accessible for this user group.

### 1. CTS-Specific Constraints

Four constraints govern all design and implementation decisions:

| Code | Constraint            | Requirement                                                                                                                                    |
|------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| C1   | Voice-First Input     | Every data entry point must offer a voice alternative. Typing is never required.                                                               |
| C2   | Minimum Touch Target  | All interactive elements are at least 88 × 88 dp. Secondary actions: 88 × 56 dp. Exceeds WCAG 2.5.5 (44 × 44 px).                              |
| C3   | No Sustained Gestures | No swipe-to-delete, drag-and-drop, long-press, or pinch as primary interactions. All actions completable with a single tap or voice.           |
| C4   | Hardware-Flexible     | Full functionality via external keyboard, switch access, and OS voice control (iOS Voice Control, Android Voice Access, Windows Voice Access). |

### 2. Large Readable Text

Font: DM Sans (Google Fonts). Chosen for round letterforms and consistent weight range (400–800). Minimum body text size is 15 sp, exceeding WCAG 1.4.4. Line height is at least 1.5× for all body content (WCAG 1.4.12). Text scales to 200% without loss of content or horizontal scrolling (WCAG 1.4.10 Reflow).

| Style               | Size  | Weight | Use                                |
|---------------------|-------|--------|------------------------------------|
| Display / Hero      | 40 sp | 800    | Landing Page headline              |
| H1 - Screen Title   | 28 sp | 700    | Dashboard, top-of-screen names     |
| H2 - Section Header | 22 sp | 700    | Card sections, list groupings      |
| H3 - Card Title     | 18 sp | 600    | Medication name, appointment title |
| Body Large          | 17 sp | 400    | Instructions, critical information |
| Body                | 15 sp | 400    | Default content (minimum)          |

|         |       |     |                             |
|---------|-------|-----|-----------------------------|
| Label   | 13 sp | 600 | Button text, nav tab labels |
| Caption | 12 sp | 400 | Timestamps, metadata only   |

### 3. Color & Contrast

All text and UI components meet WCAG 2.1 Level AA contrast ratios (4.5:1 for normal text, 3:1 for large text). Color is never the sole indicator of meaning - status labels always pair color with an icon and text.

| Color           | Hex     | Used For                     | Contrast vs White |
|-----------------|---------|------------------------------|-------------------|
| Teal Primary    | #0F6E56 | Buttons, active nav, headers | 5.74 : 1          |
| Text Primary    | #111827 | Body text, headings          | 17.5 : 1          |
| Amber Secondary | #F0A832 | Reminders, Health Log        | 6.8 : 1           |
| Red Emergency   | #E24B4A | SOS button, emergency screen | 4.6 : 1           |
| Text Secondary  | #6B7280 | Captions, metadata           | 4.6 : 1           |

### 4. Touch Friendly Buttons

Undersized targets require precise aiming, forcing CTS users to tense their wrists and triggering pain flares. All interactive elements meet or exceed 88 dp in height. Primary buttons span full screen width to eliminate aiming entirely.

- Primary buttons (Confirm Taken, Voice Log): Full width × 88 dp
- Secondary buttons (Manual Log, Skip): 88 × 56 dp
- Bottom navigation tabs: flex width × 64 dp (full screen width divided by 5)
- List rows and card tap areas: Full width × 72 dp minimum
- Form inputs: Full width × 88 dp (voice replaces typing where possible)
- SOS Emergency button: 160 × 160 dp circular - activatable under maximum distress
- Minimum 8 dp gap between all adjacent interactive elements (WCAG 2.5.8)

### 5. Screen Reader Considerations

Target screen readers: VoiceOver (iOS/macOS), TalkBack (Android), NVDA / JAWS (Windows), ARIA (Web).

- All icons have aria-label or alt text. Decorative icons are marked aria-hidden.
- Interactive elements announce context, not just label. Example: "Confirm Taken, Lisinopril 10 mg, button."
- Status changes announced via aria-live="polite". Emergency alerts use aria-live="assertive".

- Form inputs have visible labels programmatically linked via htmlFor / labelFor. Errors linked via aria-describedby.
- Heading hierarchy maintained: one H1 per screen, H2 for sections, H3 for cards.
- Modal dialogs trap focus and return focus to the trigger element on close.

Bottom nav announces: "Home, selected, tab 1 of 5."

## 6. Simple Navigation

- Maximum 2-tap depth: every primary feature is reachable in 2 taps from Dashboard.
- Persistent bottom navigation bar (5 tabs) always visible on main screens. No hidden drawer menus.
- SOS always 1 tap away: Emergency tab is always visible in the bottom nav from any screen.
- Back navigation uses a visible Back button. Swipe-back is available but never the only option.
- Consistent layout: App Bar always top (avatar left, title center, actions right). Bottom nav always bottom.
- No time limits or auto-advancing content (WCAG 2.2.1).

## 7. Voice & Alternative Input

Every workflow can be completed without typing. Voice input is provided at three levels:

- In-app voice: Voice Log button on Dashboard and Health Log activates native speech-to-text. Platform APIs: iOS Speech Framework, Android SpeechRecognizer, Web Speech API.
- OS-level voice control: All interactive elements carry accessible labels discoverable by iOS Voice Control, Android Voice Access, and Windows Voice Access.

External keyboard: Full tab-order navigation. Focus indicator is a 3 px teal outline at 3:1 contrast ratio. All actions reachable without touch.